BENJAMIN GRIMSLEY

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PROFESSIONAL SUMMARY

Accomplished Senior Systems Engineer & Technical Leader with expertise in Model-Based Systems Engineering (MBSE), interface control, requirements development, and aerospace/defense system architecture. Proven track record delivering complex missile defense, lunar lander, GN&C, VLEO, and autonomous systems for DARPA, US Navy, USAF, and commercial space programs. Skilled in SysML, functional decomposition, modeling and simulation, CI-integrated verification, trade studies, risk management, and Python/NLP automation for requirements traceability. Adept at collaborating across engineering, operations, and government stakeholders to drive mission success.

WORK HISTORY

Jul 2023 – Present ***Senior Member of Technical Staff, Systems Engineering****, Draper, Cambridge MA*

Defense Space Program | *Interface Control Manager*

- Own avionics specification and ICD development across guidance, avionics, power, thermal, communications, ground launch, and host vehicle interfaces, under configuration control in 3DEXPERIENCE.
- Model and simulate the system against the government reference architecture, assessing how the design aligns with government-defined functions and interfaces.
- Decompose system functions from L3 to L4 and allocate them to subsystems, connecting the government reference architecture to the subsystem interfaces captured in the ICDs.

Trident II D5LE2 Mk8 Guidance System | *MBSE Lead, Integration & Test (US Navy)*

- Lead MBSE for the Mk8 integration and test cells, building an ISO/IEC/IEEE 42010 architecture description in SysML on a four-view methodology (operational, behavioral, structural/interface, traceability) to support requirements-to-design gap analysis ahead of System Functional Review (SFR); authored and presented the SFR methodology brief.
- Built stakeholder, actor, and reusable use case models with specifications for four priority test cells, modeling unit-under-test interfaces from the ICD as typed proxy ports; self-taught Cameo DataHub to sync requirements.
- Found inverted traceability semantics in an inherited model, set the corrective relationship pattern, and completed the refactor, starting it inside the SFR window to keep technical debt from compounding.
- Wrote the use case specification for a flight avionics hardware-in-the-loop (HWIL) facility delivered to the prime contractor, structured around actors, stakeholders, and traceability to drive requirements sessions with the HWIL team.
- Authored and led the Mk8 Guidance System CONOPS through 26-reviewer approval, closing 144 comments and establishing lifecycle use cases that drove requirements, interfaces, and functional/behavioral modeling.

Guidance-System Simulation | *Requirements & Verification*

- Rebuilt a 32-line informal requirements list into a specification aligned with ISO/IEC/IEEE 29148 and the INCOSE Guide for Writing Requirements, leading review rounds with the task and simulation leads that took 127 candidate requirements to 82, each recording its origin and tracing to a 58-item use case catalogue or stating why not.
- Wrote the requirements to be verifiable inside the team's GitLab CI pipeline, keeping contractual verification method separate from automatability and assigning per-merge, nightly, release, or milestone gates to 42 of the 82.
- Mapped the existing 47-job pipeline to the requirements it actually evidenced, finding that most analysis jobs produced plots without asserting anything and the one compliance check was deliberately allowed to fail without blocking; wrote the requirement making per-merge checks blocking unless a waiver is recorded.
- Defined a staged approach to judging model fidelity before real-system reference data exists (conformance to the system model, analytic reference cases, conservation and dimensional invariants, and a no-degradation ratchet against an accepted baseline), deferring accuracy thresholds to a stated data-availability trigger.
- Built automated consistency checks for the requirements workbook (two-way referential integrity, text-to-register agreement, near-duplicate detection, revision deltas), each proven on a planted defect; drafted the V&V plan.

Blue Origin Lunar Lander & On-Orbit Refueling

- Served as modeling liaison for on-orbit refueling and propellant transfer, coordinating architecture and system models across stakeholders; built lunar lander models, requirements analysis, and mission simulation for NASA reviews.
- Built a Python NLP pipeline tracing requirements from JAMA to V&V activities modeled in Cameo, combining BM25, sentence-transformer similarity, domain vocabulary, and query expansion to rank explained candidate matches for engineer review; traced 758 requirements against 395 activities, cutting manual tracing by an estimated 30%.
- Added an INCOSE-rule quality grader with gap detection that surfaced 166 orphans and 1,482 quality issues, plus an evaluation harness (precision/recall/F1@k, MRR, Hit@k) scoring each revision against ground truth.
- Took over antenna modeling effort stalled 10 weeks; unblocked power budget analysis for the NASA design review.

- Delivered requirements gap analysis presented at Blue Origin's company-wide Requirements Summit, prompting adoption of JAMA-based system function–to–requirement traceability to surface quality gaps earlier.

DARPA Otter VLEO Satellite & Conventional Prompt Strike

- Led requirements analysis for the DARPA Otter VLEO satellite, developing 200+ traceable, testable system and subsystem requirements; lauded by DARPA and Redwire for a robust framework at System Requirements Review.
- Developed trade studies and models for M-Code GPS receiver integration on the Conventional Prompt Strike hypersonic weapon (signal integrity, interfaces, mission constraints); successful flight tests Dec 2024 and May 2025.

Jun 2021 – Jun 2023 ***Technical Project Manager****, Boston Engineering Corporation, Waltham MA*

- Engaged US Navy autonomous vehicle community, developed user stories, and captured use cases in SysML for automated anchoring software built on ROS2; delivered anchoring management software on time and on budget.
- Led 14 engineers in developing an automated QA/QC system for thermal spray coating on Navy flight decks, using ROS2 for platform-agnostic, real-time ISO roughness evaluation; completed on time and $125K under budget.
- Supervised a multidisciplinary team and managed a $2.6M R&D portfolio for America's warfighters, owning requirements, trade studies, and functional analyses.
- Led model development and analysis that cut the dry weight of an explosive ordnance disposal tether-management ROV by 60%, eliminating the need for a human technician in a mine threat area.

Jun 2019 – Jun 2021 ***Captain, Program Manager, Integrated Base Defense****; Hanscom AFB, MA*

- Managed 3 acquisition programs for 60+ Air Force stakeholders; owned QA, cost, schedule, and performance.
- Led $783M contract; evaluated 31 proposals for Force Protection Electronic Security, awarded 4 months early.

Jun 2018 – Jun 2019 ***1st Lieutenant, Launch Integration Program Manager****; Vandenberg SFB, CA*

- Led maintenance team supporting booster stacking for Minotaur I/IV rockets on east and west coast ranges.
- Led vehicle stacking for NASA's Orion Ascent-Abort 2; oversaw rocket processing and range prep for NROL-129.

Jun 2016 – Jun 2018 ***2nd Lieutenant, Lead Facilities Systems Engineer****; Vandenberg SFB, CA*

- Supervised 3-member team overseeing $70M+ in modernization and sustainment of critical space launch facilities.
- Led $622K HVAC project and managed schedule risk; closed unit's #1 readiness issue for spaceplane recovery.

PROJECTS

St. Lawrence ShipWall | *Personal project* · github.com/bsullgrim/shipwall

- Built a ship-traffic display for a family cottage on the Seaway, where async Python services on a Raspberry Pi ingest AISStream's WebSocket feed, detect channel crossings through a shared state machine, and drive a 128×64 HUB75 LED matrix via an ESP32-S3, with a browser mock as the executable service-to-firmware interface spec.
- Operate it unattended and remotely over Tailscale with a three-layer watchdog (firmware heartbeat, serial watchdog, systemd backstop), a gated deploy script, hourly off-device backups, and a numbered incident log with write-ups on 20 failures, from ext4 corruption after a power loss to a lost system clock breaking TLS.

The Holliston Ledger | *Founder and Editor, 2026 – Present* · hollistonledger.com

- Launched an independent civic news publication on Holliston, MA town government in October 2026 and run it single-handedly, producing a Town Meeting voter guide, a recurring briefing and *This Week at Town Hall* wrap-up, and a hub page that pulls 88 scattered town filings on a Chapter 40B housing proposal into one readable account.
- Built a Python pipeline that finds unminuted recordings across a dozen boards, transcribes them with WhisperX and pyannote diarization tuned to local names, and flags every motion, vote, dollar figure and recusal to check against the video, feeding a Google Sheets review queue via GitHub Actions so nothing publishes without a human check.

EDUCATION & CERTIFICATIONS

Master of Science, Systems Engineering — Johns Hopkins University (2024)
Bachelor of Science, Mechanical Engineering — Rochester Institute of Technology (2016)
Program Management Level II — Defense Acquisition University (2017) | **Security Clearance: TS/SCI** (Active)

SKILLS & AFFILIATIONS

MBSE & Tools: SysML, Cameo/MagicDraw, Cameo DataHub, 3DEXPERIENCE, JAMA, DOORS, VTL/Report Wizard
Programming: Python (spaCy, Sentence Transformers, BM25), MATLAB, Groovy, Git, GitLab CI, AI-assisted workflows
Systems Engineering: ICDs, Functional Decomposition, M&S, Traceability, V&V, Trade Studies, Risk Management
Standards: ISO/IEC/IEEE 29148, ISO/IEC/IEEE 42010, INCOSE Guide for Writing Requirements
Domains: Missile Defense, Lunar Landers, Hypersonics, Satellite/VLEO, GN&C, Launch Systems, Defense Acquisition
Affiliations: INCOSE – PMI – ASME – National Eagle Scout Association – Air Force Officers Association